
UNIT 22 INTERTEMPORAL OPTIMISATION-I

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22.0 OBJECTIVES

After going through this unit, you should be able to:

- distinguish between static and dynamic models as well as between continuous and discrete dynamic models
- enumerate various dynamic optimization approaches, state the essential idea of each approach, state the theoretical results which form the foundations of the approach, and use each one for solving dynamic optimisation problems. More explicitly,
- explain the Calculus of Variations approach, state the foundational theorem justifying the basis of the approach, and use it in solving appropriate problems.
- Explain the dynamic programming approach for dynamic optimization, and using it derive optimality criteria of discrete dynamic optimisation.

22.1 INTRODUCTION

Note: In this unit, a number of theorems are stated and proved. The proofs may be skipped altogether, provided the essential underlying ideas become clear from the statements of the theorems. One should go through the proofs only for getting clarifications about the underlying essential ideas.

Modern study of economic theories entails study of behaviour of economic agent/s who tend to optimise her desired outcome at the end of process. One of the fundamental principles of mathematical economics, is that, economic agent/s try to maximize an objective function subject to resource constraints. This principle is adopted to explain economic behaviour at one instance of time as well as the evolution of economic behaviour through time. Optimisation from an economist's sake can be broadly classified into static optimisation and dynamic optimisation. In contrast to static optimisation, in dynamic optimisation, the variables of model are time variant or in other words they change with respect to time. Each of the various Dynamic optimisation techniques consists of set of tools which can be useful, apart from in Economics, in operations research, engineering and other applied field; moreover, these tools have applications in different subfields including economic growth macroeconomics, microeconomics, finance, and dynamic games. Main methods for solving Dynamic optimisation problems can depend on (i) Pontryagin's maximum principle by Lev. S. Pontryagin and (ii) Dynamic programming principle by Richard Bellman. Pontryagin's method is based on ideas from classical calculus of variations whereas Bellman's principle is based on recursive method which is akin to the concept of sub-game perfect equilibrium in context of dynamic game theory. Dynamic programming is not to be misconstrued as synonym of dynamic optimisation rather it is one of the methods to solve optimal control problems.

Dynamic optimisation problem can be divided into two major categories: discrete and continuous. This distinction is based upon the nature of objective function whether that is continuous or piecewise continuous. Moreover, the objective function can be deterministic or stochastic in nature. The study of stochastic dynamic optimisation is beyond the scope of the syllabus. We shall first consider the unconstrained optimisation and then delve into constrained optimisation. But even before that, we need to have clarification between some closely-connected pairs of concepts.

22.1.1 Some Conceptual Clarifications

The two pairs of concepts discussed are (i) static optimization vs. dynamic optimization, and (ii) function vs. functional. Also, the concept of *differentiability class* is defined.

22.1.1.1 Static Optimization vs. Dynamic Optimization

We briefly illustrate the essential difference in the subject-matter of the current block, viz. *dynamic optimization*, vis-à-vis the subject-matter of the previous two blocks, which we retrospectively call *static optimization*. For this purpose, we consider a very simplified case of *profit maximization*. Consider the case of a producer of, say *fruit-icecream* using three fruits, viz. mango, grapes and banana.

Let

$CP(\text{fruit}) = (CP(\text{mango}), CP(\text{grape}), CP(\text{banana}))$, the cost-price-vector per unit of the fruits,

$Q(\text{fruit}) = (Q(\text{mango}), Q(\text{grape}), Q(\text{banana}))$, the quantity vector used per unit of icecream,

$SP(\text{Icecream})$ = sales-price per unit of icecream

Total-Production (Icecream) = the total production of fruit-icecream

Thus,

Total-revenue = Total-Production (Icecream) $\times$ $SP(\text{Icecream})$.

Total-cost = Total-Production (Icecream) $\times$ $\{CP(\text{fruit}) \cdot Q(\text{fruit})^T\}$, where Q^T is the transpose of matrix Q , and ' $\cdot$ ' denotes product of two matrices.

Total-Profit = *Total-revenue* – *Total-cost*, is the **objective function**, the value of which is sought to be optimized, subject to $CP(x)$, $Q(x)$ etc.

In the above, the variables $CP(x)$, $Q(x)$, $SP(\text{Icecream})$, Total-Production (Icecream) are treated as independent variables.

However, on deeper analysis, we find that the prices of fruits depend on the season, i. e. the **time** period of the year. For example, *mango, in India*, is the cheapest (and, of course, of the best quality) during peak period, viz. from May to August. And its price varies over the year according to the distance in time from the peak period. Similarly, grape, in India, has the peak period from January to May. Thus, $CP(x)$ is **not independent variable, but dependent on time t** .

Hence, regarding **price**, **instead of $CP(x)$, its appropriate modelling is by $CP(x, t)$** . Similarly, the **quantities**, of various fruits used in one unit of icecream, depends on time, e. g., during February-April, more of grape, and less of mango may be used because of respective availability and quality during the period. Hence, **instead of $Q(x)$, the appropriate modelling is by $Q(x, t)$** .

Also, regarding **sales**, the units of icecream sold depends on the time of the year, e. g. during the summer period of April to July, sales may be at the peak, while during December-February, the sales may be the least. Thus, **Total-Revenue**, and the objective function **Total-profit** also are functions of time.

Thus, in such problems, **time** is an important independent variable, and in many of such cases, **the only** independent variable. Other variables, like $CP(x)$, $Q(x)$,

and objective function *Total-profit*, each is a function of t , the time. Such problems are modelled better by **Dynamic optimization** which essentially involves **time** as one of the independent variables, and generally the only independent variable.

22.1.1.2 Function vs. Functional

We have already discussed, in some detail the concept of *function*, in Unit 3 of Block1.

A function $f: X \rightarrow Y$ is a rule which associates to each element of X , a unique element of Y . Generally, the set $\mathbb{R}^n$, for $n \geq 1$, is taken as the domain X ; and $\mathbb{R}$, the set of real numbers, as the codomain Y .

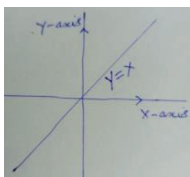
Let us now consider a **set of functions**, say,

Set of Functions = $\{f, g, h$, where each is a function from $\mathbb{R}$ to $\mathbb{R}$, such that

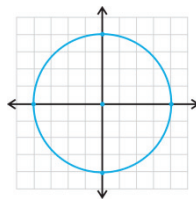
i) $y = g(x) = x$;

ii) $y = h(x) = +\sqrt{(b^2 - x^2)}$ and (iii) $y = k(x) = x^2$ }.

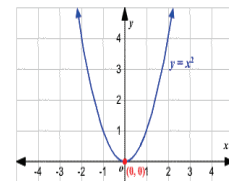
The three functions are graphically represented as



$$g(x) = x$$



$$h(x) = +\sqrt{(b^2 - x^2)}$$



$$k(x) = x^2$$

Next, we define the following functions from *Set of Functions* to $\mathbb{R}$

- I) $F_{l, b}$: Set of Functions $\rightarrow \mathbb{R}$, such that for $y \in \text{Set of Functions}$,
 $F_{l, b}(y) = \text{length of the curve represented by } y = f(x) \text{ from } x=0 \text{ to } x=b$.

From Integral Calculus, we know that the length of the arc of the curve $y=f(x)$ included between two points with $x=a \geq 0$ and $x=b \geq 0$ is given by

$$\int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \int_a^b \sqrt{1 + f'^2(x)} dx$$

In order to calculate the required lengths for the three given functions, first we calculate $dg/dx = 1$, and $dh/dx = (-x)/\sqrt{(b^2 - x^2)}$, and $dk/dx = 2x$

By taking $a=0$, we get

i) $F_{l, b}(g) = \int_0^b \sqrt{1 + 1} dx = (\sqrt{2}) b$;

ii) $F_{l, b}(h) = (1/4) \text{ circumference of circle of radius } b = (\pi b)/2 \text{ (directly without explicitly using integral)}$

iii) $F_{l, b}(k) = \int_0^b \sqrt{1 + (2x)^2} dx = \dots$

II)

$F_{area, b}$: Set of Functions $\rightarrow \mathbb{R}$, such that for $y \in \text{Set of Functions}$, $F_{area, b}(y) = \text{area of the curve represented by } y=f(x) \text{ from } x=0 \text{ to } x=b > 0, \text{ and } y > 0$.

From Integral Calculus, or otherwise, we know that the required area under the curve is given by

i) $F_{area, b}(g) = (1/2) b^2$

ii) $F_{area, b}(h) = (1/4) \pi b^2$

iii) For the next area, we know from Integral Calculus that the area of the curve $y=f(x)$ included between two points with $x=a \geq 0$, $x=b \geq 0$, and $y > 0$ is given by

$$\int_a^b f(x) dx. \text{ Therefore, the required area is}$$

$$F_{area, b}(k) = \int_0^b x^2 dx = b^3/3.$$

Leaving aside the details under (I) and (II) above, the essential idea of the discussion is that the INPUTS, to each of the two functions $F_{l, b}$ and $F_{area, b}$, ARE (the whole) functions themselves, e. g. $h(x) = +\sqrt{(b^2 - x^2)}$. And the values returned by each of $F_{l, b}$ and $F_{area, b}$ are associated to the whole function, and not to a point on the function.

Such functions to which the inputs are functions are called FUNCTIONALS. Functionals play significant role in dynamic optimization.

Another concept which we encounter frequently in Dynamic Optimization is that of **Differentiability Class**, which we introduce next.

22.1.1.3 Differentiability Class

Let U be an open subset of $\mathbb{R}$, the set of real numbers, and $k \geq 0$, be an integer. Then a function $f : U \rightarrow \mathbb{R}$ is said to be of **differentiability class C^k** , if the first k derivatives f' , f'' , ..., and $f^{(k)}$ exist and are continuous on U . In this case, the term '*f has smoothness of degree k*' is also used sometimes. Obviously, $C^{k+1} \subseteq C^k$, for all k . The function f is said to be infinitely differentiable, smooth, or of class C^∞ , if it has derivatives of all orders on U .

22.2 SOME EXAMPLES

We start this section with providing you some examples of continuous dynamic optimisation and discrete dynamic optimisation.

22.2.1 Example of Continuous Dynamic Optimisation

Consider a profit making firm which wishes to maximize its profit over a period of time. Let the beginning of time of calculation be t_0 , and end time be t_f . The current profit depends on both current output and rate of change of output with

respect to time. If $x(t)$ denotes output at time t , then the rate of change of output is $x'(t)$, and $V(x(t), x'(t), t)$ denotes the profit functional. Thus, the problem may be mathematically modeled/expressed as

$$\text{Max} \int_{t_0}^{t_f} V(x(t), x'(t), t) dt. \quad \dots\dots\dots(22.1)$$

$$\text{Subject to } x(t) \geq 0 \quad \quad \quad x(t_0) = x_0$$

22.2.2 Example of Discrete Dynamic Optimisation

The Cake Eating Problem

Assume that there is a cake whose size at time t is denoted by W_t and a muncher who wants to eat it in T time periods. Let C_t denote the size of cake eaten during time-periods $(t, t+1)$. The initial size of the cake is $W_0 = \alpha$ (say), and finally, we have $W_T = 0$. The eater has psychological discount factor $0 < \beta < 1$ and a static logarithmic utility function. The question here is what should be the optimal eating strategy.

The problem is to find the optimal path $C^* = \{C_t^*\}_{t=0}^{T-1}$ and $W^* = \{W_t^*\}_{t=0}^T$ that solve the problem

$$\text{Max} \sum_{t=0}^T \beta^t l_n(C_t) \dots\dots\dots (22.2) \text{ subject to } W_{t+1} = W_t - C_t;$$

$$W_0 = \alpha, \text{ and } W_T = 0$$

For handling the type of problems discussed above through the two examples, there are mainly three dynamic optimization approaches, viz. (i) calculus of variations, (ii) dynamic programming, and (iii) optimal control theory, which is modern generalization of variational calculus. We start our discussion with the first of these.

22.3 THE CALCULUS OF VARIATIONS

The origin of calculus of variations can be traced back to John Bernoulli who in year 1696 posed a problem called brachistochrone problem: (If a small object moves under the influence of gravity which path between two fixed points enables it to make the trip in the shortest time). The problem was solved by him and independently by his brother James in 1697. Other eminent mathematicians who contributed to this branch of calculus include Euler, Lagrange, Legendre, Jacobi, Hamilton, Weierstrass, Hilbert.

Next, we discuss fundamental result, of Calculus of Variations, viz, ***Euler-Lagrange Equation, also known as Euler-Lagrange Condition, or even E-L condition, EL equation etc.***)

22.3.1 Theorem for Optimality

Theorem: Let $J: C^2[x_0, x_f] \rightarrow \mathbb{R}$ be a functional

$$J(x(t), t) = \int_{t_0}^{t_f} V(x(t), x'(t), t) dt \quad \dots\dots\dots(22.3)$$

Where $V(.)$ has continuous partial derivative of second order with respect to $x(t)$, $x'(t)$ and t and $x_{t_0} = x_0$ and $x_{t_f} = x_f$ and $x_{t_0}, x_{t_f} \in \mathbb{R}$, $t_f > t_0$. Then there exists an external path x^* for which the objective function $J(.)$ is either maximized or minimized, if and only if.

$$\frac{\partial V(.)}{\partial x(t)} - \frac{d}{dt} \frac{\partial V(.)}{\partial x'(t)} = 0 \quad \dots\dots\dots(22.4)$$

The above equation is a second order (generally non linear) ordinary differential equation that any continuous and differentiable external x^* must satisfy. This differential equation is called the Euler-Lagrange equation. The boundary values associated with this equation for the fixed end point problem are $x(t_0) = x_0$, $x(t_f) = x_f$

Euler-Lagrange equation is necessary condition if $x(t)$ is single variable we get 2nd order differential equation but if $x(t)$ is a $n \times 1$ vector i.e. consists of n variables, then we get $2n$ differential equations. In general these equations are non linear time varying equations.

Proof: We are assuming that t_0 and t_f are fixed and they are the end points.

If the point A corresponds to t_0 and B to t_f in Fig. 22.1 then let $x^*(t)$ be optimal path from A to B and this $x^*(t)$ will optimise (either maximize or minimize) functional $J(.)$

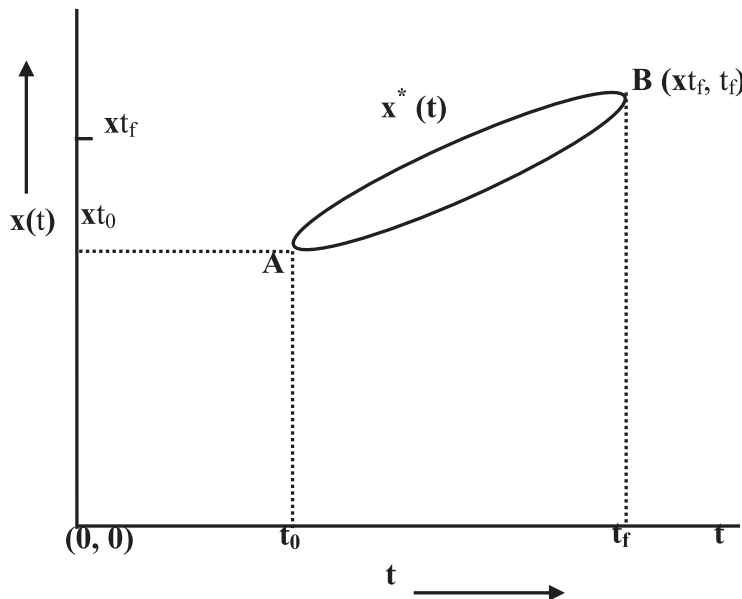


Fig. 22.1: Representation of Optimal and Sub-Optimal Path

Let $x_a(t)$ is a path very close to optimal path but it is sub-optimal path:
Let $x_a(t) = x^*(t) + \delta x(t)$ (22.5)

$\|\delta x(t)\|_2 > 0$ and satisfy following boundary condition

$$x_a(t_0) = x^*(t_0) = x_0$$

$$x_a(t_f) = x^*(t_f) = x_f$$

$\delta x(t)|_{t=t_0} = 0, \delta x(t)|_{t=t_f} = 0$, that means A and B are fixed points.

Lemma:- Suppose that a function $g(t)$ is continuous on interval t_0 to t_f then

$$\int_{t_0}^{t_f} g(t) \delta x(t) \cdot dt = 0 \quad \dots(22.6)$$

if and only if

$$g(t) = 0 \text{ everywhere over the interval } t_0 \text{ to } t_f$$

Our task is to find a trajectory (path) $x^x(t)$ connecting (t_0, x_0) and (t_f, x_f) along which $J(x(t), x'(t), t)$ is extremum.

Using the same analogy as we have used in static optimization i.e., $\nabla f(x) = 0$, where x is a vector of n variables.

We can write

$\delta J =$ First variation of functional

$$= \int_{t_0}^{t_f} \left\{ \left[\frac{\partial V(\cdot)}{\partial x(t)} \right]_* \cdot \delta x(t) + \left[\frac{\partial V(\cdot)}{\partial x'(t)} \right]_* \cdot \delta x'(t) \right\} dt = 0 \quad \dots(22.7)$$

$$\text{For expanding the term } \int_{t_0}^{t_f} \left[\frac{\partial V(\cdot)}{\partial x'(t)} \right]_* \delta x'(t) dt \text{ we use} \quad \dots(22.8)$$

Integration by parts rule

$$\text{We know } d(UV) = Udv + VdU$$

Integrating both side within limit t_0 to t_f

$$\int_{t_0}^{t_f} d(Uv) = U \int_{t_0}^{t_f} dv + v \int_{t_0}^{t_f} du$$

$$[Uv]_{t_0}^{t_f} = u \int_{t_0}^{t_f} dv + v \int_{t_0}^{t_f} du$$

$$\int_{t_0}^{t_f} u dv = [Uv]_{t_0}^{t_f} - \int_{t_0}^{t_f} v du$$

$$\text{We can write } \delta x'(t) = \frac{\partial}{\partial t} \delta x(t)$$

$$\text{Taking } \frac{\partial V(\cdot)}{\partial x'(t)} \text{ as } u \text{ and } d x(t) \text{ as } d v$$

Therefore, we can write equation (22.8) as

$$= \left\{ \left[\frac{\partial V(\cdot)}{\partial x'(t)} \right]_* \cdot \delta x(t) \right\}_{t_0}^{t_f} - \int_{t_0}^{t_f} \delta x(t) \cdot \frac{d}{dt} \left(\left[\frac{\partial V(\cdot)}{\partial x'(t)} \right]_* \right) \cdot dt \quad \dots(22.9)$$

Since both ends are fixed

$$\delta x(t) \text{ at } t = t_0 \text{ is zero and } \delta x(t) \text{ at } t = t_f \text{ is zero.}$$

Now we derive the necessary condition for functional to optimise is δJ (1st variation of functional) = 0. The Sufficient condition for functional to optimise is $\delta^2 J$ (2nd variation of functional) > 0 for minimum and $\delta^2 J < 0$ for maximum value of functional.

Because of our assumption that $v(\cdot)$ is continuous and twice differentiable by using Leibniz theorem we can differentiate $v(\cdot)$ within integral.

Hence $J(x_a(t), x'_a(t), t)$

$$= J(x^*(t) + \delta x(t), x'^*(t) + \delta x'(t), t)$$

$$= J(x^*(t), x'^*(t), t) + \Delta J \text{ (incremental functional value)}$$

Using Taylor series expansion around $x^*(t)$

$$\Delta J =$$

$$\int_{t_0}^{t_f} \left\{ \left[\frac{\partial V(\cdot)}{\partial x(t)} \right] \delta x(t) + \left[\frac{\partial V(\cdot)}{\partial x'(t)} \right] \delta x'(t) \right\}_{\substack{x(t)=x^*(t) \\ x'(t)=x'^*(t)}} + \frac{1}{2} \left[\frac{\partial^2 V(\cdot)}{\partial x^2(t)} \right] (\delta x(t))^2 + \left[2 \frac{\partial^2 V(\cdot)}{\partial x(t) \partial x'(t)} \right] \delta x(t) \delta x'(t) + \left[\frac{\partial^2 V(\cdot)}{\partial x'^2(t)} \right] (\delta x'(t))^2 \Big|_{\substack{x(t)=x^*(t) \\ x'(t)=x'^*(t)}} + \text{other higher order terms} \Big\} dt \dots (22.10)$$

For sake of brevity, we henceforth will use the terminology $*$ as $x(t) = x^*(t)$
 $x'(t) = x'^*(t)$ $\Delta J \approx \delta J + \delta^2 J$ (ignoring higher order term) $\delta^2 J$ is the terms which are multiplied by $\frac{1}{2}$ in above equation.

Because both end points are fixed.

$$\left\{ \left[\frac{\partial V(\cdot)}{\partial x'(t)} \right]_* \cdot \delta x(t) \right\}_{t_0}^{t_f} = 0$$

$$\text{Hence, we can write 22.8 as } \int_{t_0}^{t_f} \left[\frac{\partial V(\cdot)}{\partial x'(t)} \right]_* \cdot \delta x'(t) dt = \int_{t_0}^{t_f} \left[\frac{\partial V(\cdot)}{\partial x(t)} \right]_* \cdot \delta x(t) dt \dots (22.11)$$

Therefore, we can write (22.7) as

$$\delta J = \int_{t_0}^{t_f} \left\{ \left[\frac{\partial V(\cdot)}{\partial x(t)} \right] - \frac{d}{dt} \left[\frac{\partial V(\cdot)}{\partial x'(t)} \right] \right\} \delta x(t) \cdot dt = 0$$

recalling our lemma $\int_{t_0}^{t_f} g(t) \delta x(t) dt = 0$ iff $g(t)$ which is a continuous function is zero at every point over the interval to t_0 to t_f . Using the lemma we get necessary condition for extremum

$$\left[\frac{\partial V(\cdot)}{\partial x(t)} \right] - \frac{d}{dt} \left[\frac{\partial V(\cdot)}{\partial x'(t)} \right]_* = 0 \quad (\text{same as 22.4})$$

When $x(t)$ is $n \times 1$ vector the necessary condition is

$$\left[\frac{\partial V(\cdot)}{\partial x(t)} \right] - \frac{d}{dt} \left[\frac{\partial V(\cdot)}{\partial x'(t)} \right]_* = 0_{n \times 1} \dots (22.12)$$

This equation is known as Euler-Lagrange equation. The curve $x^*(t) = x(t, C_1, C_2)$

C_1 and C_2 are integration constant and can be found by initial condition.